

IPv6

Objectives

After completing this lesson, you should be able to:

- Explain the background for IPv6 development
- Describe some features of IPv6
- Explain the IPv6 address format
- List and describe IPv6 address types
- Describe aspects of IPv6 link-local unicast, global unicast, and multicast address layouts
- List the three types of autoconfiguration for obtaining IPv6 addresses
- Explain Stateless Address Autoconfiguration (SLAAC)
- Explain two IPv6 settings for specific interfaces
- Describe some IPv6 commands



IPv6: Overview

- This is the most recent version of the IP specification.
- In the 1990s, the Internet Engineering Task Force (IETF) spearheaded efforts to address a pending IP address shortage.
- Classful addressing resulted in inefficient address allocation.
- Efforts were made to mitigate the IP address issue:
 - Classless Inter-Domain Routing (CIDR) with Variable-Length Subnet Masking (VLSM)
 - Network Address Translation (NAT)
- RFC 8200 is the current IPv6 specification—July 2019:
 - Succeeds IPv4, though both coexist currently
 - Addresses the IPv4 address shortage
 - Provides new functionality

IPv6: Features

- Increased address space
- Autoconfiguration
- Header is fixed length with fewer fields—less processing overhead
- Extension headers can be specified to support various options
- Flow label
- Scope

IPv6 Addresses

- IPv6 Addresses use 128 bits.
- The first part of the address is a prefix:
 - Analogous to the network portion of an IPv4 address
 - Specified with the form: prefix/prefix-length-in-bits
 - Contents of the prefix depend on the address type
- The second part of the address is the interface identifier (ID), which can be set in various ways:
 - Derived from the system's MAC address—modified EUI-64 format
 - Randomly generated to protect privacy
 - Manually
- Autoconfiguration simplifies address assignment.

IPv6 Address Representation

- 128-bit IPv6 addresses contain eight sets of 4 hexadecimal numbers delimited by colons (each number is 4 bits), for example:

```
2001:0db8:0000:0000:0a00:20ff:feb5:4139
```

- Guideline is to use the shortest representation possible:

- Leading 0s (zeros) are removed, giving:

```
2001:db8:0:0:a00:20ff:feb5:4139
```

- Consecutive 16-bit sets of zeros are represented with a double colon (: :). You can do this only once in any address:

```
2001:db8::a00:20ff:feb5:4139
```

- The double colon must not be used in place of only one 16-bit set of zeros.
- See RFC 5952 ("IPv6 Text Representation") for more details on address representations.

IPv6 Address Types

- IPv6 supports the following address types:
 - Unicast addresses
 - Multicast addresses
 - Anycast addresses
- Specific examples of unicast addresses are:
 - Global unicast
 - Link-local unicast
- IPv6 differs from IPv4 in that IPv6 does not provide broadcast addresses as a mechanism for communicating with other hosts on a subnet.
- The high-order bits of the prefix determine the address type.

High-Order Bits Specified for Address Types

- The high-order bits in an IPv6 address determine the address type.
- Some common address types:

Address Type	High-Order Bits	Prefix
Link-local unicast addresses	1111 1110 10	fe80::/10
Global unicast addresses	001	2000::/3
Multicast addresses	1111 1111	ff00::/8

Prefix Notation

- CIDR notation specifies the prefix length of an IPv6 address.
- An example of an address with the prefix specified is:

```
2001:db8::a00:20ff:feb5:4139/64
```

- /64 indicates that the prefix is 64 bits in length.
- 2001:db8:: is the 64-bit prefix—32-bit global unicast prefix followed by 32 bits of zeros.
- a00:20ff:feb5:4139 is the 64-bit interface identifier.

Unicast Addressing

- A unicast packet is sent to a single interface with the matching destination IPv6 address.
- Two types of unicast addresses are:
 - Link-local unicast addresses are used on a local network link.
 - Global unicast addresses (also called aggregatable global unicast addresses) are routable across the Internet.

Link-Local Unicast Address Layout

- The following represents the link-local unicast address layout:

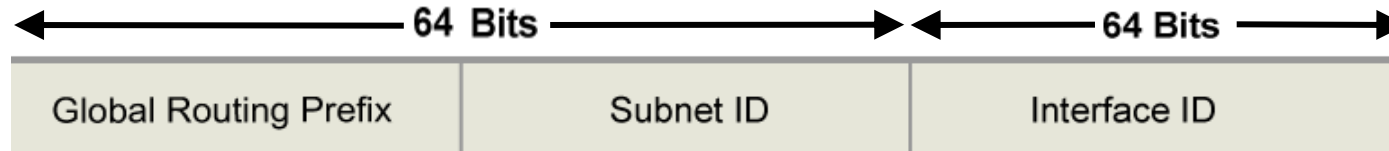
10 Bits	54 Bits	64 Bits
111111010	All Zeros (0)	Interface ID

- The first 10 bits of the address prefix identify an address as a link-local address:
 - 1111 1110 10 in binary
 - fe8 in hexadecimal
- The remainder of the prefix is padded with 54 bits of zeros and is followed by the 64-bit Interface ID.
- Example (compressed):

```
fe80::a00:20ff:feb5:4139
```
- The prefix is fe80:: and the Interface ID is a00:20ff:feb5:4139.
- Link-local addresses are not forwarded by routers.

Global Unicast Address Layout

- The following represents the global unicast address layout:



- The first three bits are always set to 001 (binary), 0x2 (hexadecimal)
- Global routing prefix—size is variable.
 - Value is assigned by an address registry or ISP to identify a site.
- Subnet ID—size is dependent on the size of the Global Routing Prefix.
 - Identifier of a subnet or link within a site, assigned by a system administrator
- Global routing prefix size + subnet ID size = 64 bits
- Interface ID—uniquely identifies an interface and is 64 bits in size
- Example with a 48-bit Global Routing Prefix, 16-bit Subnet ID, and 64-bit Interface

Global Routing Prefix	Subnet ID	Interface ID
2001:db8:0:6	a00:20ff:feb5	4139

Multicast Addressing

- A packet that is addressed to a multicast address is delivered to all the nodes that belong to the multicast group.
- It allows many systems to be members of a multicast group, which is represented by a single multicast address.
- The multicast address is a destination address and has a scope:
 - Source address is unicast.
 - Scope can range from local to global.
- Multicast addressing is also part of IPv4.

Multicast Address Layout

- The following represents the multicast address layout:

Flags		Scope	Multicast Group ID
8 Bits	4 Bits	4 Bits	112 Bits
11111111	0RPT	XXXX	

- The 8 high-order bits of a multicast address are:
 - 11111111 in binary
 - ff in hexadecimal
- A fourth flag bit ("T") of zero means the address is permanent or well known—assigned by the Internet Assigned Numbers Authority (IANA).
- Scope bits define the reach of the address, for example, 0x2 means link-local scope.
- Example of an all-routers multicast address with link-local scope:

```
ff02::2
```

Example of Multicast

Addresses	Scope	Descriptions
FF01::1	Interface-local	All nodes address
FF01::2	Interface-local	All routers address
FF01::FB	Interface-local	Multicast DNSv6
FF02::1	Link-local	All nodes address
FF02::2	Link-local	All routes address
FF02::A	Link-local	All EIGRP routers
FF02::FB	Link-local	Multicast DNSv6
FF05::2	Site-local	All routers address
FF05::C	Site-local	Variable scope allocation
FF05::FB	Site-local	Multicast DNSv6
FF0X::FB	Variable scope	Valid over all scope for Multicast DNSv6
...	...	...

Anycast Addressing

- An address is assigned to a group of interfaces.
 - These interfaces share the same anycast address.
- Packets are delivered to the nearest interface member, instead of being delivered to all members of a group.
 - Routers determine the "nearest" interface.
- Intent is to provide desired access to a service hosted on multiple nodes.
- Some potential benefits:
 - Help balance load
 - Provide quicker response time

Neighbor Discovery Protocol (NDP)

- Includes five ICMPv6 message types:
 - Router Solicitation
 - Router Advertisement
 - Neighbor Solicitation
 - Neighbor Advertisement
 - Redirect
- "Solicitation"—request for information
- "Advertisement"—response or sent independently of a solicitation
- Some Neighbor Solicitation/Advertisement purposes:
 - Discover a link-layer address, determine whether a neighbor is reachable, determine whether an address is a duplicate (Duplicate Address Detection—DAD)
- Some Router Solicitation/Advertisement information provided:
 - Routable prefixes for address autoconfiguration, default gateway, hop limit
- Redirect—routers let nodes know of a better first hop to a destination

Obtaining IPv6 Addresses

- Two general methods:
 - Manually—configure them yourself
 - Automatically with autoconfiguration
- Autoconfiguration means automatically creating or obtaining routable IPv6 addresses for network interfaces.
- Types of autoconfiguration:
 - Stateless Address Autoconfiguration (SLAAC)
 - Stateless DHCPv6—uses SLAAC for IP address and DHCPv6 for other information
 - Stateful DHCPv6—like IPv4 DHCP

Stateless Address Autoconfiguration—SLAAC

- New in IPv6
- SLAAC process begins when an interface is enabled.
- Nodes automatically generate IPv6 addresses:
 - No manual node configuration required
 - No DHCP servers necessary
- Routers do not use this method.
- No state information about addresses is maintained.
- Facilitates current and future device addressing:
 - Consider the volume of devices in the Internet of Things (IoT).

The SLAAC Process

- An interface creates a unique 64-bit interface ID in modified EUI-64 format or randomly.
- A 128-bit link-local address is generated by combining the interface ID and the 64-bit link-local prefix (`fe80::/10`).
 - At this point, this IPv6 address is considered *tentative*.
- This newly formed address is checked for duplicates on the local link—Duplicate Address Detection (DAD).
 - Neighbor Solicitation ICMPv6 message is sent to Solicited Node multicast address.
 - The system receives a Neighbor Advertisement from any device that is currently using the address.
 - If the address is already in use, autoconfiguration stops and an address must be configured manually.
 - If no reply is received, the link-local address is presumed to be unique on the local link and is assigned to the interface—the address is then *preferred*.

Interface ID—Modified EUI-64 Format Created from MAC Addresses

- Example of a 64-bit modified EUI-64 interface ID created from a 48-bit MAC address
- Example 48-bit MAC address in hexadecimal format:

08:00:20:b5:41:39

- MAC address in binary format:

0	8	0	0	2	0	b	5	4	1	3	9
0000	1000	0000	0000	0010	0000	1011	0101	0100	0001	0011	0111

- Two octets, 0xff and 0xfe, are inserted between the third and fourth octets:

0	8	0	0	2	0	f	f	f	e	b	5	4	1	3	9
0000	1000	0000	0000	0010	0000	1111	1111	1111	1110	1011	0101	0100	0001	0011	0111

Interface Identifier Creation—Modified EUI-64 Format

- Bit 9 (from the left), the universal/local bit, is toggled. This completes the process:

0	a	0	0	2	0	f	f	f	e	b	5	4	1	3	9
0000	1010	0000	0000	0010	0000	1111	1111	1111	1110	1011	0101	0100	0001	0011	0111

- The modified EUI-64 interface ID in IPv6 hexadecimal format:

0a00:20ff:feb5:4139

- This unique interface identifier is 64 bits of the 128-bit IPv6 address. It is combined with a prefix to form a complete IPv6 address.
- Example 128-bit IPv6 address formed from a global unicast (documentation) prefix and the modified EUI-64 interface ID:

2001:0db8:0000:0000:0a00:20ff:feb5:4139

- The same address compressed:

2001:db8::a00:20ff:feb5:4139

Stateless DHCPv6 Autoconfiguration

- IPv6 address is obtained via SLAAC.
- Other information is provided by a DHCPv6 server.
 - for example, DNS server address(es)
- Router Advertisement has "Other Configuration" (O flag) set to 1.
 - This tells a node to autoconfigure its IPv6 address
 - Then to request DHCPv6 information that is needed
- No state information about addresses is maintained.

Stateful DHCPv6 Autoconfiguration

- Provides greater control over addressing than SLAAC
 - Might be desired in some enterprise environments
- Requires setup of a DHCPv6 server
- Router advertisement has "Managed address configuration" (M flag) set to 1
 - Tells the node to request address and other information from a DHCPv6 server
- DHCPv6 server supplies more information like DHCP in IPv4
 - DNS Addresses, Default Gateways, Redundancy configuration and many more
- DHCP Unique Identifier (DUID) used for client and server identification rather than MAC addresses as in DHCP for IPv4
- No state information about addresses is maintained.

DHCPv6 – Comparison between Stateless and Stateful

Stateless

- **Inferior Security:** SLAAC is susceptible to attacks that involve spoofing link-layer addresses.
- **Limited control:**
- **Limited options:** Less configuration
- **Easier to setup:**

Stateful

- **Better security benefits:**
- **Suited to larger networks**
- **More control and options:** Network that needs detailed configuration and management.
- **More complex to setup:**

Other Address Types

- IPv4-mapped IPv6 addresses:
 - Carrying an IPv4 address in an IPv6 packet is called an IPv4-mapped IPv6 address.
 - This format is used for nodes that only support IPv4.
- Unspecified address:
 - The source address of a system that has not had an address assigned will be all zeros - `::` in compressed format.
 - It is analogous to the IPv4 address `0.0.0.0`.
- Loopback address:
 - `::1` in compressed format
 - This address is analogous to the `127.0.0.1` local address used by IPv4 systems.

Using IPv6 with Oracle Linux

- IPv6 is enabled in Oracle Linux 9 by default.
- Check this by looking for `inet6` addresses as follows:

```
# ip addr|grep inet6
inet6 ::1/128 scope host
inet6 fe80::2f49:669e:6cb1:2a1/64 scope link ...
inet6 fe80::d86b:9b28:2646:effb/64 scope link ...
```

- The example shows the IPv6 loopback address of `::1/128`; the scope is `host` - the node itself only.
- It also shows two link-local addresses for two different interfaces:
 - `fe80::` is the link-local prefix; the scope is `link` - the local link only.
 - Prefix length is 64 bits.
 - The interface IDs are `2f49:669e:6cb1:2a1` and `d86b:9b28:2646:effb`.

IPv6 Settings for Specific Interfaces

- IPv6 parameters can be set in ifcfg-<interface> files in /etc/sysconfig/network-scripts/ for specific interfaces, for example:
 - IPV6ADDR=<IPv6 address>[/prefix length]
 - Allows a primary static IPv6 address to be provided
 - 64 is the default prefix length if not specified
 - IPV6_ADDR_GEN_MODE=stable-privacy|eui64
 - Specifies the types of interface IDs used for creating addresses
 - "stable-privacy" enables random creation of interface IDs that do not change within the current subnet (see RFC 9219).
 - "eui64" specifies that interface IDs are configured in modified EUI-64 format from the MAC address
 - Default is eui64

IPv6 Settings for Specific Interfaces

- Enable IPV6 with dhcp method

```
# nmcli con modify ens160 ipv6.method dhcp connect.autoconnect yes
```

- Enable IPV6 with link-local

```
# nmcli con modify ens160 ipv6.method link-local
```

Enable IPV6 with static configuration

```
# nmcli con modify ens160 ipv6.method manual ipv6.address 2000:aaaa:bbbb::1/64  
ipv6.gateway 2000:aaaa:bbbb::1111 ipv6.dns 8888:8888::4444:4444
```

Some IPv6 Commands

- Linux commands specific to IPv6 can be specified using the "-6" option. For example:

- Show IPv6 addresses:

```
# ip -6 addr
```

- Ping an IPv6 address:

```
# ping -6 2001:db8::a00:20ff:feb5:4139
```

- To ping an IPv6 link-local address, the interface name must be appended to the IPv6 address with the "%" character:

```
# ping -6 fe80::e89d:a31:aecf:a5a2%eth0
```

- Or the interface name must be specified with the "-I" option:

```
# ping -6I eth0 fe80::e89d:a31:aecf:a5a2
```

Quiz



IPv6 supports which of the following address types? (Select all that apply)

- a. Unicast
- b. Multicast
- c. Anycast
- d. Broadcast



Quiz



IPv6 supports which of the following address types? (Select all that apply)

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Quiz



A unicast packet is sent to a single interface.

- a. True
- b. False



Quiz



A unicast packet is sent to a single interface.

- a. True
- b. False



Quiz



Which of the following are the three high order bits of a Global Unicast Address?

- a. 100
- b. 010
- c. 001
- d. 000



Quiz



Which of the following are the three high order bits of a Global Unicast Address?

- a. 100
- b. 010**
- c. 001
- d. 000



Summary

In this lesson, you should have learned how to:

- Explain the background for IPv6 development
- Describe some features of IPv6
- Explain the IPv6 address format
- List and describe IPv6 address types
- Describe aspects of IPv6 link-local unicast, global unicast, and multicast address layouts
- List the three types of autoconfiguration for obtaining IPv6 addresses
- Explain Stateless Address Autoconfiguration (SLAAC)
- Explain two IPv6 settings for specific interfaces
- Describe some IPv6 commands



Practices for Lesson 14: Overview

This practice covers using IPv6.